

Harnessing R Packages for Modern Artificial Intelligence

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Agenda

- Introduction (LLM)
- Learning Objective
- R packages for LLM
 - API Integration
 - R packages
- Use Cases
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- Questions

Introduction

Artificial Intelligence especially Large Language Models (LLMs) is rapidly reshaping how we analyze data, write code, and build tools in R.

This workshop introduces you to powerful tools through hands-on exercises designed to show how AI can work *with you* inside R, not outside it.

Learning Objective

By the end of this workshop, participants will be able to:

- a. Understand what LLMs can do inside R
- b. Use R packages designed for AI workflows
- c. Build small, functional AI-powered tools
- d. Apply AI to real data science workflows

R packages for LLM

- ellmer
- gander
- Btw
- mall
- Shinychat
- mcptools
- ragnar
- kuzco
- buggy
- Ensure
- chores

- **ellmer** – Lets you chat with LLMs (like GPT or Claude) directly from R for text generation, coding help, and analysis
- **gander** – An AI assistant inside RStudio that reads your code context and helps refactor, explain, or fix it.
- **btw** – Sends R objects (like data frames) as context to LLMs so they can analyze or summarize them.
- **mall** – Performs row-wise NLP tasks (like sentiment or classification) across data frame text columns. ([nlptask](#))
- **shinychat** – Adds an interactive AI-powered chatbot panel into Shiny applications.

- **kuzco** – Provides AI-based computer vision capabilities such as describing or analyzing images.
- **vitals** – Uses AI to evaluate model predictions, performance, and classification outputs.
- **ragnar** – Enables Retrieval-Augmented Generation (RAG) in R by letting LLMs answer questions using your documents.
- **mcptools** – mcptools connects R to AI tools using the Model Context Protocol, it lets AI tools talk to R sessions, and lets R talk to external tools like GitHub or Google Drive.
- **chores** – Automatically writes documentation, tests, and code improvements using AI. (Mostly used by R developers and package authors)
- **buggy** – Detects bugs and errors in R code using AI-powered debugging.
- **ensure** – Generates unit tests automatically for your R functions using AI.

R packages for LLM API Integration

What is API?

- API (Application Programming Interface) is a structured way for your R code to *talk to* an external service.
- When you use ellmer or other LLM packages, your R session sends:
 - A request (your prompt)
 - To an AI model provider (e.g., OpenAI, Anthropic)
 - And receives a response (text, JSON, structured output)
- APIs allow R to use powerful cloud-based models without installing them locally.

Models

LLM packages in R typically allow you to choose from different types of models:

- Chat models – conversation-style (GPT-4, Claude, Gemini)
- Text models – long-form generation or summarization
- Embedding models – convert text into numeric vectors
- Vision models – describe images (for kuzco)
- Function/tool calling models – execute structured tasks

Each model has different strengths (reasoning, speed, cost, multimodal handling).

Model Providers

R packages interact with major AI companies offering LLMs:

- OpenAI: GPT-4, GPT-4o, GPT-3.5, Embeddings
- Anthropic: Claude 3/Opus/Sonnet/Haiku
- Google (Gemini): Gemini 1.5/1.0 models, Vision models
- Mistral AI: Mixtral series, Codestral
- Cohere: Command R family
- Open-source models via APIs: Llama-3, Qwen, Mistral (served on providers like Fireworks, Replicate, Groq)

Most R packages let you switch providers by changing a single parameter or environment variable.



Questions??

Let's Jump into some hands on and Use cases